universität wien
Faculty of Psychology

Lernen unter COVID-19-Bedingungen

Pupils

Method Report

1. Introduction

The competence for self-regulated learning (SRL) are considered important determinants of learning achievements, well-being, and ultimately health (e.g., Schober et al., 2013; Wirth & Leutner, 2008). In the wake of the COVID-19 pandemic, pupils worldwide had to switch to home learning more or less unprepared and faced a fundamentally altered situation where self-regulated learning has become a necessity. The present cross-sectional study aims at investigating the relevance of SRL for well-being and learning success, both quantitatively and qualitatively, in a large sample consisting of Austrian secondary school pupils, at three measurement points in the first half of 2020.

2. Study Description

2.1. Title and version

Lernen unter COVID-19-Bedingungen: SchülerInnen (Learning under COVID-19 conditions: pupils, V1.0)

2.2. Principal Investigators

Univ.-Prof. Dr. Barbara Schober, Department of Developmental and Educational Psychology, University of Vienna
Ass.-Prof. Dr. Marko Lüftenegger, Department of Developmental and Educational Psychology, University of Vienna
Univ.-Prof. Dr. Dr. Christiane Spiel, Department of Developmental and Educational Psychology, University of Vienna

2.3. Funding/Acknowledgments

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2.4. Fieldwork agency

The fieldwork of the study “Lernen unter COVID-19-Bedingungen” was managed and handled internally.

2.5. Topics/Keywords

- Sociodemographic information: age, sex, gender-role orientation, education, language, region
- Situation at home: household structure, help provided, accessible electronic devices
- Crisis perception: perceived challenges, information distribution, school attendance, communication about crisis
- Effects of the crisis: changes in well-being, learning, motivation, social relatedness, competence, autonomy, and self-regulated learning

- Learning: basic needs (social relatedness, competence, autonomy), self-regulated learning, learning goals, learning time, learning environment, metacognition, procrastination, flow, perseverance, learning motivation
- Education: achievement, satisfaction, importance of grades
- Online-education: digital learning, digital teaching, student-teacher-contact, hours doing schoolwork

3. Study Design

3.1. Sampling

The sample of the study "Lernen unter COVID-19-Bedingungen" consisted of Austrian pupils attending secondary school with access to the internet. The pupils were invited to participate in the survey by their teachers, headmasters, and by calls in radio and television programs. Moreover, the link to and a recommendation to participate in the survey was published by the Federal Ministry of Education, Science, and Research on their website. The link with a respective invitation to the study was also placed on the survey's website. In order to participate in the study, the respondents had to agree to the informed consent, be Austrian residents, and attend secondary school. It is important to note that the sample cannot be considered representative for the Austrian population as many students do not have easy access to the internet and/or devices needed to fill out the survey. Therefore, risk groups may be underrepresented in the current sample.

3.2. Fieldwork

The overall data collection was conducted in the time from 01 April 2020 to 29 June 2020.

- Wave 1: 02 April 2020 – 24 April 2020
- Wave 2: 27 April 2020 – 15 May 2020
- Wave 3: 08 June 2020 – 29 June 2020

4. Data Cleaning

4.1. General data cleaning procedure

Cleaning procedure was the same for all three data sets collected in three measurement points and contained two steps:

4.1.1. Initial data cleaning

As participating students answered an online questionnaire via Unipark (Questback, 2007), data was exported from unipark in .csv format and imported to SPSS. Variables names were replaced. All missing values were defined as -99.

Invalid zero values or values of -77 were replaced with -99. Participants who gave consent, answered affirmatively to questions end1 („Ich habe den Fragebogen sorgfältig ausgefüllt“) and end2 („Ich habe die Fragen ehrlich beantwortet“), and had answering time (variable duration) larger than zero were selected for a new SPSS file.

4.1.2. Data cleaning

As unipark assigns -1 to variable duration in cases where the break was made during the answering of the questionnaire, all valid -1 cases were replaced with (+) 9999999 value. For all open-ended questions that needed to be cleaned a new variable was made with suffix .clean in the variable name and there all non-integer values were replaced with a valid integer value (e.g., mean values of stated inputs, see the rules for cleaning open-ended questions below), all string values were converted to numeric (see below) and all invalid inputs were replaced with -99 (see example below). Participants with 100% missings on main outcome variables (positive affect, social relatedness, competence, autonomy, SRL scales and procrastination) were excluded. Minimum possible duration was tested by filling in the questionnaire as fast as possible while reading all items. All cases above the duration threshold and within a defined age-range (based on data from Statistics Austria) were selected for a new SPSS file.

Rules for cleaning open ended variables in measurement point 1

Variables ap.study.norm („Wie viele Stunden am Tag machst du normalerweise Dinge für die Schule außerhalb des Unterrichts?“) and ap.study.curr („Wie viele Stunden am Tag machst du derzeit Dinge für die Schule?“)

- when it is written “*from-to*” change to average of these two numbers, e.g., 3-4, 3 bis 4, 3 4 = 3.5
- when three numbers stated in a row, take the mean value, e.g., 4-5-6 = 5
- commas in number always need to be replaces with dot, e.g., 3,5 = 3.5
- inputs in minutes need to be changed into hours, e.g., 20min = 0.33
  (however, inputs like 1min or 2min need to be replaces with -99)
- fractions should be treated same as “*from-to*” cases, e.g., 3/4 = (as 3-4) 3.5, 1 1/2 = 1.5, 1-1 ½ = 1.25
- in case of additional symbols next to the whole number, will be treated as “from - to” cases e.g., < 1 = 1 → 0-1 = 0.5, 1+ = 1.5, > 1 = 1 → 1-2 = 1.5, 3-4+ = 3-4.5 = 3.75
- approximately symbol should only be deleted, as well as “ca.”, “bis zu”, “mind”, “max”, “min” in front of the number ~ 4 = 4, ca. 3 = 3, bis zu 2 = 2, Max 2 = 2, mind. 3 = 3
- Cases with colon will be treated as hours’ indicators, while dot remain as dot 1:30 = 1.5 = 1h30 = 1.5, 1.30 = 1.3
- Assign missing to all strange ranges and unclear inputs, e.g., 2-36 = -99, rund = -99

Variable ap.persons („Wie viele Personen, außer dir, leben bei euch zu Hause?“)

- Assign missing (-99) to all unclear inputs, e.g., -1 = -99, 1 (Wohnsitz 1) / 5 (Wohnsitz 2) = -99
- Where possible and clear, count or summarize the number of persons in household, e.g., 11 Kinder + 3 Betreuer = 14, 3 Kinder 2 Erwachsene = 5
- All cases where stated that kid lives alone, asssign missing, e.g., Niemand = -99
- When more situation/living settings are stated, always take the current one, e.g., aktuell 4, normalerweise 3 = 4

Variable **ap.kids** („Wie viele davon, außer dir, sind Kinder, die zur Schule gehen?“)

- Assign missing (-99) to all unclear inputs, e.g., 1.5 = -99
- When it is written “insgesamt”, always -1 to get the number of kids besides the participant, e.g., Insgesamt zwei Kinder = 1
- Where possible and clear, count or summarize the number of kids in household, e.g., Mein Bruder, aber er bleibt auch gerade daheim und lernt dort. = 1
- Kids in Kindergarten and Vorschule do not count, University students do not count as kids that go to school, Students in Berufschule, as well as Lehre do count

In measurement point 2 and 3 these variables were in form of closed questions.

4.1.3. Missing values

All empty spaces were replaced with -99. Missing values that were represented by 0 in the raw data set are replaced by -99 on all variables where 0 is not a valid value. Missing values represented with -77 in unipark were replaced with -99 as well.

4.1.4. Filters

Data records of participants who did not give their consent are deleted. Furthermore, data records of participants who did not indicate that they answered diligently and honestly are deleted.

A fastest possible time for answering the questionnaire is determined diligently. Variable **duration** of answers with duration below the fastest possible time is set to 9999999.

Data records with 100% missings on main outcome variables are deleted.

4.1.5. Answers to open questions

Answers to open questions in measurement point 1 were cleaned (see Rules for cleaning open ended variables above). For time variables two versions are kept with one containing all original values (**ap.study.norm.orig**, **ap.study.curr.orig**), and the second containing values with a cut off on 12 hours (**ap.study.norm**, **ap.study.curr**). Original values that go beyond 12 hours are set as missing in the latter version. In measurement point 2 and 3 these variables were in the form of closed questions.

4.1.6. Multiple choice / single choice

Question „Welche Sprache(n) sprichst du zu Hause?“ in all three measurement points and “Welche Schulart besuchst du?” in the first measurement point were posed in the multiple choice format.

4.2. Anonymisation procedure

4.2.1. Gender variable

In the third measurement point (cp.gender) answers indicating that the corresponding person chose “divers” were recoded to missing as there were less than 20 observations.

4.2.2. Language variables

Data on the languages spoken were gathered via multiple choice and an open answer box for additional languages that were not eligible in the multiple choice question. The information from the open answers is reduced to a dichotomous variable (“yes”/“no”) that indicates if additional languages were named. Answers indicating “no”, unclear inputs and missing answers are set as “no”. In the third measurement point answers indicating that the corresponding persons speaks Persian (cp.lang.per), Polish (cp.lang.pol) or Slovenian (cp.lang.slo) are set to missing as each of these variables has less than 20 observations.

4.2.3. Open questions

In measurement point 1 ten open variables are deleted without recoding or standardizing the answers:

- ap.school.other (“Welche Schulart besuchst du? Andere”)
- ap.grade (“In welche Klasse gehst du?”)
- ap.who (“Falls du derzeit zu Hause Hilfe beim Lernen bekommst, wer hilft dir vor allem? Jemand anderer”)
- ap.concerns (“Was bereitet dir derzeit die größten Sorgen?”)
- ap.pos (“Was findest du derzeit besonders schön?“)
- ap.chall (“Was ist beim Lernen derzeit besonders schwierig?”)
- ap.coping (“Was machst du derzeit, wenn dir beim Lernen etwas schwer fällt?”)
- ap.coping.succ (“…und wie gut gelingt dir das?”?
- ap.study.succ (“Was gelingt dir beim Lernen derzeit besonders gut?”)
- ap.support (“Wobei würdest du derzeit beim Lernen Hilfe brauchen?”).

In measurement point 2 six open variables are deleted without recoding or standardizing the answers:

- bp.who (“Falls du derzeit zu Hause Hilfe beim Lernen bekommst, wer hilft dir vor allem? Andere”)
- bp.info.more (“Worüber würdest du mehr Information brauchen?”)
- bp.studya.better (“Wenn sich etwas verbessert hat: Was ist das genau?”)
- bp.studya.worse (“Wenn sich etwas verschlechtert hat: Was ist das genau?”)
- bp.dont.miss (“Was vermisst du gar nicht?”)
- bp.miss.most (“Was vermisst du am meisten?”).

In measurement point 3 thirteen open variables are deleted without recoding or standardizing the answers:

- cp.who (“Falls du derzeit zu Hause Hilfe beim Lernen bekommst, wer hilft dir vor allem? Jemand anderer”)
- cp.no.school.curr.change (“Hat sich für dich etwas verändert, seitdem deine Mitschüler*innen in die Schule gehen?”)
- cp.school.curr.like (“Was findest du derzeit in der Schule besonders schön?”)
- cp.school.curr.dontlike (“Was gefällt dir derzeit in der Schule gar nicht?”)
- cp.study.better (“Wenn sich etwas verbessert hat: Was ist das genau?”)
- cp.study.worse (“Wenn sich etwas verschlechtert hat: Was ist das genau?”)
- cp.dont.miss (“Was vermisst du gar nicht?”)
- cp.miss.most (“Was vermisst du am meisten?”)
- cp.hl.experience (“Was nimmst Du aus der Zeit des Home-Learnings für dich mit?”)
- cp.hl.hobbies (“Hast du während der Zeit zu Hause neue Dinge für dich entdeckt (z. B. neue Hobbies, Sportarten, ...)?”)
- cp.hl.future.worries (“Wenn du an den kommenden Herbst denkst, worüber machst du dir am meisten Gedanken?”)
- cp.no.school.curr.other2 (“In welcher Form hast du derzeit Kontakt mit deinen Lehrer*innen? Anderes”)
- v_418 (“Wo verbringst du die Hausaufgabentage? Anderes”).

4.2.4. Open questions containing numeric values

Answers to questions requiring numeric values are recoded according to the description in section "Data cleaning procedure". Original (string) answers are deleted.

4.2.5. Age variable

Data containing the age of the participants in measurement points 1 and 2 are categorized by truncating decimal places in original values to get groups with at least 20 observations.

4.2.6. School type variables

Values containing "Hauptschule (HS)" or "Mittelschule (MS)" are merged to one category ("Hauptschule (HS) / Mittelschule (MS)") in each measurement point as they are technically equivalent with "Hauptschule" being a deprecated label for "Mittelschule". Furthermore, in measurement point 2 values indicating “Sonderschule” as school type are recoded into “Andere”. This provides groups with at least 20 observations.

4.3. Categorizing variables

Variables containing numeric values in connection with time and numbers of people were categorized into suitable categories that allow analyses among different points of measurement.

4.4. Further deletions

4.4.1. Consent variable

One variable containing the participants' agreement on processing the filled in data for scientific purposes is deleted in each measurement point (ap.consent, bp.consent, cp.consent) as all included participants gave their consent for processing the filled in data for scientific purposes.

4.4.2. Filter variables

One filter variable (filter_$) originating from the cleaning procedure is deleted in each measurement point.

4.4.3. Missing values only

In measurement point 3, the numerical variable cp.anticipation ("Wie sehr vermisst du es, in die Schule zu gehen?") was deleted as the question did not appear in the online questionnaire due to technical issues.

4.4.4. System variables

System variables that contain information of the duration of answering separate pages are deleted.

Measurement point 1:

- date_of_first_mail (date of first e-mail)
- datetime (date of first access)
- date_of_last_access (date of last access)
- rts5450640, rts5450641, rts5450642, rts5450643, rts5450644, rts5450645, rts5455865, rts5450646, rts5450647, rts5450648, rts5455962, rts5450649, rts5452239 (relative timestamps for pages)

Measurement point 2:

- date_of_first_mail (date of first e-mail)
- datetime (date of first access)
- date_of_last_access (date of last access)
- v_357 (captcha question)
- v_358 (expected answer to captcha question)
- captcha_answer (answer to captcha question)
- captcha_answer_numeric (numeric answer to captcha question)
- rts5481344, rts5481347, rts5481352, rts5481358, rts5481723, rts5482064, rts5482065, rts5482067, rts5482068, rts5482069, rts5482072, rts5483258, rts5483259, rts5483881, rts5483882, rts5483883, rts5484004, rts5498625 (relative timestamps for pages)

Measurement point 3:

- date_of_first_mail (date of first e-mail)
- datetime (date of first access)
- date_of_last_access (date of last access)
- v_357 (captcha question)
- v_358 (expected answer to captcha question)
- captcha_answer (answer to captcha question)
- captcha_answer_numeric (numeric answer to captcha question)
- rts4388180, rts4388183, rts4388184, rts4388186, rts4388188, rts4388189, rts4388190, rts4388191, rts4388192, rts4388193, rts4388194, rts4388195, rts4388198, rts4388199, rts4388200, rts4388218, rts4388513, rts4389566, rts4400712, rts4400713, rts4404081, rts4404082 (relative timestamps for pages)

4.5. Increasing reusability

To increase the reusability of the data set variables are rearranged into an appropriate order. Variable names are renamed to get more meaningful names.